

Determinants

Linear transformation

Two by two, three by three matrices were doing the task of linear transformation.

It squeezes the things in or stretches the things out.

Determinants

The tool to measure exactly how much things are stretched out or squeezed in.

Example:

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

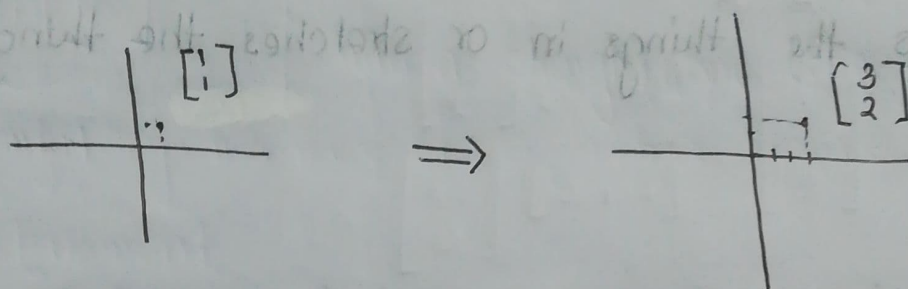
scales the $\hat{i}$ by a factor of 3

$$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

scales the $\hat{j}$ by a factor of 2

If we multiply the vector with the matrix we obtain the transformed matrix.

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$



The area of the vector has been stretched out being multiplied with the matrix

Length = 1

Width = 1

Area = 1

Length = 3

Width = 2

Area = 6

The matrix $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ has scaled the area of the vector by a factor of 6

* The very special factor by which linear transformation changes any area is the determinant.

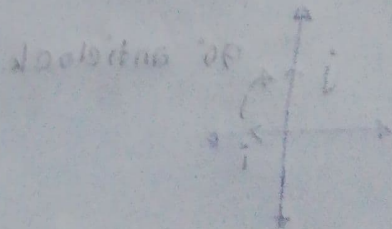
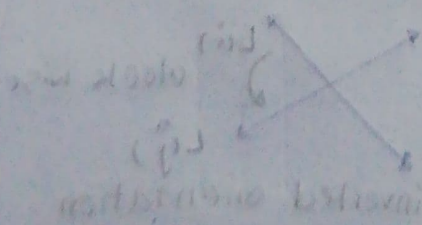
$\det \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix} = 2$, if we multiply any particular vector by $\begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$, the area of vector will be stretched out by the factor of 2.

$\det \begin{pmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix} = \frac{1}{2}$, if we multiply any particular matrix by $\begin{pmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{pmatrix}$, the area of the particular vector will be squeezed into half.

Exceptional Cases

1. $\det(A) = 0$

2. $\det(A) < 0$



$$\underline{\det(A) = 0}$$

for example, $\det \left(\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \right) = 0$

Why?

$$\begin{bmatrix} \boxed{4} & \boxed{2} \\ \boxed{2} & \boxed{1} \end{bmatrix}$$

dependent

So, the linear transformation done by this matrix will be squeezing it into a straight line (Plane $\rightarrow$ Straight line/point)

A plane is squeezed into a straight (line/point) which has no area $\therefore \text{area} = 0$

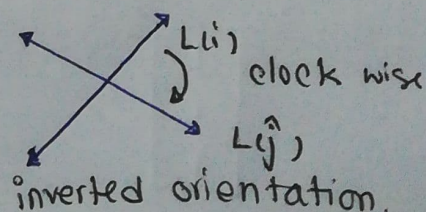
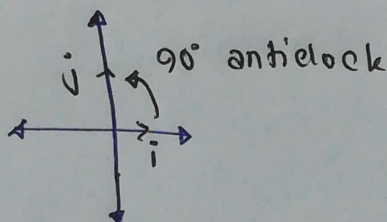
$$\underline{\det(A) < 0}$$

(negative)

Has something to do with orientation

$$\det \left(\begin{bmatrix} 1 & 1 \\ -1 & -3 \end{bmatrix} \right) = -5,$$

Here the orientation of space is inverted, so the determinant will be negative



In 2D,

Determinants work with area,

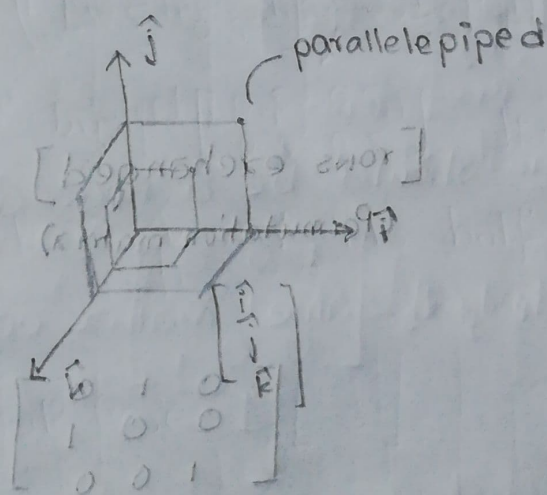
How much the area has been squeezed in or stretched out.

In 3D,

Determinants work with the idea of volume,

How much volume has been stretched/squeezed (scaled),

In 3D Space,



-ve determinant in 3D plane refers to change in orientation of the basis vectors

○ determinant squeezes a volume into point/line

Properties of Determinants (Major 3)

① $\det(I) = 1$

Determinant of Identity matrix is 1.

$$\det \begin{pmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{pmatrix} = 1, \quad \det \begin{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{pmatrix} = 1$$

② Exchanging any two rows of a matrix reverse the sign of determinant

We know, $\det(I) = 1$

$$\det(P) = 1 \text{ or } -1$$

[rows exchanged]
(Permutation matrix)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

I

$$\det(I) = 1$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

P_{21}

$$\det(P_{21}) = -1$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

P_{23}

$$\det(P_{23}) = 1$$

III All about linear combination

(a) Multiplying any row by a scalar factor has the following effect.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \xrightarrow{tr_i} \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} \rightarrow t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\det = ad - bc$$

$$\begin{bmatrix} 0 & d & 0 \\ 0 & d & 0 \\ t & a & b \end{bmatrix} = A$$

$$\det = t(ad - bc)$$

(b) Adding something with a row has the following effect

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

This linearity is only applicable for the rows, not for the whole matrix.

$$\therefore \det(A+B) \neq \det(A) + \det(B)$$

iv) If any two rows are equal, then the det. is 0. (property 2)

$$A = \begin{bmatrix} a & b & c \\ a & b & c \\ d & e & f \end{bmatrix} \quad \det(A) = x$$

$$A' = \begin{bmatrix} a & b & c \\ d & e & f \\ a & b & c \end{bmatrix}$$

$$\det(A') = -x$$

(as rows were exchanged,

$= A$

As rows were exchanged, there should be a neg. determinant. But as the matrix is unchanged, same matrix can't have different determinant values

So, only when $x = 0$, its sign doesn't matter

$$(a+b) + (a) + (b) \neq (a+a) + (b+b) \therefore$$

(v) While elimination, if $l \times \text{row}_1$ is subtracted from row k , the determinant doesn't change. (iii)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{r_2 = r_2 - l \times r_1} \begin{bmatrix} a & b \\ c - la & d - lb \end{bmatrix} \quad (v) + (b)$$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \begin{vmatrix} a & b \\ -la & -lb \end{vmatrix} \quad (iii) b$$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} + l \times \begin{vmatrix} a & b \\ a & b \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad (iii) a$$

$$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} + l \cdot 0 \quad (iv)$$

(vi) Complete row of zeros lead to $\det(A) = 0$

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\text{if } t=0, \begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} = 0 \quad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 0$$

(VII) For any upper triangular matrix, the determinant is the product of the diagonal elements.

$$\det(U) = \begin{vmatrix} d_1 & x & x \\ 0 & d_2 & x \\ 0 & 0 & d_3 \end{vmatrix} = d_1 \cdot d_2 \cdot d_3$$

Reasoning:

$$U = \begin{bmatrix} d_1 & x & x \\ 0 & d_2 & x \\ 0 & 0 & d_3 \end{bmatrix} \rightarrow \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}$$

ref. form

after elimination

$$= d_1 d_2 d_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

I

$$= d_1 d_2 d_3 \left[\det(I) = 1 \right]$$

$$0 = \begin{vmatrix} d & 0 \\ 0 & 0 \end{vmatrix} = 0 = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

(VIII) Combining (VI) and (VII) we can say (IX)

$$\det(A) = 0 \text{ when } A \text{ is singular.}$$

$$\det(A) \neq 0 \text{ when } A \text{ is invertible.}$$

Whenever A is singular, it will not have a pivot in any particular row / column. So, the whole row of that matrix will be 0, so, its determinant will be zero.

Again, $\det(A) \neq 0$ when A is invertible

Example:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{r_2 = r_2 - \frac{c}{a}r_1} \begin{bmatrix} a & b \\ c - \frac{c}{a}a & d - \frac{c}{a}b \end{bmatrix}$$

$$\rightarrow U = \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \end{bmatrix}$$

$$\det(U) = a \cdot \left(d - \frac{bc}{a}\right) = a \cdot d - b \cdot c$$

*det $\rightarrow$ find U and then multiply diagonals.

$$\textcircled{IX} \det(AB) = (\det A) (\det B)$$

$$\det(A+B) \neq (\det A) + (\det B)$$

$$\det(AB) = \det(A) \det(B)$$

→ Helps to determine the determinant of any A^{-1} ?

$$AA^{-1} = I$$

$$\Rightarrow \det(AA^{-1}) = \det(I)$$

$$\Rightarrow \det(A) \cdot \det(A^{-1}) = 1$$

$$\Rightarrow \boxed{\det(A^{-1}) = \frac{1}{\det(A)}}$$

$[A]$ has to be invertible
 $\therefore \det(A) \neq 0$

Example

$$A = \begin{bmatrix} 1 & 6 \\ 2 & 11 \end{bmatrix}$$

$$\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{11 - 12} = -1$$

$$(*) \quad \det(A^T) = \det(A)$$

Homework

$$A = LU$$

$$\det(LU) = \det(L) \cdot \det(U)$$

$$\Rightarrow \det(A) = 1$$

[In this case, L has all elements along principal diagonal = 1]

$$\Rightarrow \det(A) = 2$$

$$A^T = (LU)^T$$

$$\Rightarrow A^T = U^T L^T$$

$$\Rightarrow \det(A^T) = \det(U^T) \cdot \det(L^T)$$

$$= \det(U^T) \cdot 1$$

[diagonal elements of L are unchanged]

$$= 2 \cdot 1$$

$$= 2$$

$$\therefore \boxed{\det(A) = \det(A^T)}$$

* Cramer's Rule

Homework

1. What is $\det(A^2)$?

$$\det(A^2) = \det(A, A)$$

$$= \det(A) \cdot \det(A) \quad [\text{Property (ix)}]$$

$$= [\det(A)]^2$$

$$\therefore \det(A^2) = \{\det(A)\}^2$$

$$\boxed{\det(A) = \det(A)}$$

2. What is $\det(2A)$? [Distribution, not linear/row-wise]

$$\det, 2A = 2 \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$
$$= \begin{bmatrix} 2a & 2b & 2c \\ 2d & 2e & 2f \\ 2g & 2h & 2i \end{bmatrix}$$

$$\text{Now, } \det(2A) = \det \left(\begin{bmatrix} 2a & 2b & 2c \\ 2d & 2e & 2f \\ 2g & 2h & 2i \end{bmatrix} \right)$$

$$= \begin{vmatrix} 2a & 2b & 2c \\ 2d & 2e & 2f \\ 2g & 2h & 2i \end{vmatrix}$$

$$= 2 \cdot 2 \cdot 2 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$\underbrace{\hspace{10em}}_A$

[2 from each row, $\textcircled{\text{III a}}$]

$$= 2^3 \det(A)$$

$$\therefore \det(2A) = 2^3 \det(A)$$